Response of Some Haematological Parameters in Adult Male Wistar Rats Fed Different Colour Seeded *Zea mays L* (Maize)

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ABSTRACT

Maize (Zea mays L.) is a globally consumed staple, with its nutritional composition—particularly carotenoid content—varying by kernel color. These differences may influence blood-related health markers, yet the haematological effects of coloured maize varieties remain unclear. This study investigated the influence of white and yellow maize-based diets on selected haematological parameters in adult male Wistar rats. Twenty-five rats were randomly assigned to five groups; a control fed standard rat chow while groups 2 and 3 received diets containing 25% and 50% white seeded maize and groups 4 and 5 received 25% and 50% yellow seeded-maize for 21 days. Blood samples were collected post-treatment and analyzed for packed cell volume (PCV), hemoglobin (Hb), red blood cell count (RBC), white blood cell count (WBC), and platelet count using an automated hematology analyzer and the data analysed for statistical differences using ANOVA at $p<0.05$. There was a significant reductions in PCV and RBC in the groups fed 50% white seeded-maize and 25% and 50% yellow-seeded maize compared with control. However, Hb, WBC, and platelet counts remained unaffected across all groups. These findings suggest anemic-inducing potential of high maize consumption; particularly yellow seeded-maize. Thus, high maize content diet-especially yellow seeded-maize, may impair PCV and RBC, though immune and clotting functions appear unaffected in rats. Further research is needed to assess the nutrient profiles of these maize varieties and explore dietary strategies to counter balance any negative effects.

Keywords: Zea mays, Haematological parameters, Different colour seeded, Rats.

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1. INTRODUCTION

Maize (*Zea mays L.*) is a fundamental staple crop worldwide, providing a crucial source of carbohydrates, proteins, and micronutrients for both humans and livestock [1]. The nutritional makeup of maize varies considerably among its varieties, with distinct phytochemical compositions observed between different colored types. Notably, white and yellow maize differ in their carotenoid content; yellow maize is abundant in carotenoids like beta-carotene-a vitamin A precursor-as well as lutein and zeaxanthin, which support eye health [2]. In contrast, white maize lacks these carotenoids but may possess other valuable nutrients. Such differences in phytochemical and nutrient profiles among maize varieties could affect the physiological and hematological responses of animals.

Haematological indices play a vital role in evaluating an animal's overall health and nutritional condition. These parameters reflect the functional status of the blood and bone marrow, the primary site of blood cell formation [3]. Important haematological measurements include packed cell volume (PCV), haemoglobin (Hb) concentration, red blood cell (RBC) count, white blood cell (WBC) count, and platelet count.

PCV, also referred to as haematocrit, represents the percentage of blood volume

composed of red blood cells and serves as a critical marker for diagnosing anaemia or polycythaemia. Haemoglobin, the oxygen-carrying protein within red blood cells, indicates the blood's capacity to transport oxygen. The red blood cell count measures the number of RBCs per unit volume of blood. Variations in PCV, Hb concentration, and RBC count are commonly assessed together to diagnose disorders affecting oxygen transport and red blood cell production, which can be impacted by dietary and nutritional factors [4].

WBCs, or leukocytes, serve as the body's primary defense against infections and foreign substances. An elevated WBC count typically reflects inflammation or infection, whereas a reduced count may suggest immunosuppression [5]. Thus, monitoring WBC levels offers valuable insight into an animal's immune status. Platelets, also known as thrombocytes, play a critical role in blood clotting and maintaining haemostasis. Alterations in platelet numbers can impair clotting ability, potentially resulting in bleeding disorders or thrombosis [6].

Numerous studies have examined how different dietary components influence haematological parameters across various animal species. For instance, certain diets rich in phytochemicals exhibit antioxidant properties that protect blood cells from oxidative stress, thereby enhancing haematological health [7]. In contrast, nutrient deficiencies often lead to conditions such as anaemia or compromised immune function. Considering the nutritional disparities between white and yellow maize especially in carotenoid content and antioxidant potential, it is reasonable to hypothesize that these maize varieties may induce distinct haematological effects in Wistar rats. Consequently, a comparative investigation is warranted to elucidate how the phytochemical and nutritional differences in these maize types impact blood cell production, immune responses, and overall haematological status in adult male Wistar rats.

Despite being a staple food for a large portion of the global population, there is a significant gap in the literature concerning the specific effects of different colored maize varieties on hematological parameters. Although the nutritional and phytochemical distinctions between white and yellow maize-particularly the presence of carotenoids such as beta-carotene in yellow maize are well established [2], the direct physiological impacts of these differences on blood composition remain inadequately understood.

Previous studies have shown that dietary factors can markedly influence hematological indices including PCV, Hb, and RBC count [4]. However, comparative investigations assessing the effects of white versus yellow maize on these parameters within controlled animal models are limited. Most existing research either evaluates a single maize variety or focuses on alternative feed types, hence a critical knowledge gap.

This paucity of data presents a challenge, as without clear insight into how different maize varieties affect hematological profiles, it is difficult to make evidence-based nutritional recommendations for animal feeding and potentially human health. For example, if one maize variety consistently improves hematological outcomes, its promotion in livestock diets could enhance animal health and productivity. Conversely, if certain varieties negatively influence these parameters, appropriate dietary interventions could be designed to counteract such effects. Additionally, the influence of maize color on WBC and platelet counts-which relate to immune function and coagulation-remains underexplored, leaving important questions unanswered regarding immune response and blood clotting in animals fed these diets. The present study aimed to assess the impact of ingesting different colours seeded maize on some haematological parameters in adult male Wistar rats. The specific objectives include to estimate PCV, Hb, RBC, WBC and platelet

count in adult Wistar rats fed white or yellow seeded maize.

2. MATERIALS AND METHODS

Collection and identification of materials: Two different coloured seeded maize/corn (white seeded maize; believed to originate from UK and yellow seeded maize; believed to originate from USA) grown in Nigeria, were purchased from Ekpoma local market in Edo State. Rat chow (Bendel feeds and flour meal Ewu, Nigeria) was obtained from a local store in Ekpoma. The materials and equipment used in this study include; plastic animal cages with top wire meshed, feeding troughs and drinkers, Electronic weighting scale, micropipette, plain tubes, sample and Ethylene Diamine Tetra Acetic acid (EDTA) bottles, synthetic hand gloves, wooden dissecting board, dissecting sets, syringes and needles, cotton wool, filter paper, measuring cylinder, beakers, disinfectants and detergents (for washing and cleaning), ice block, cooler, wooden shadings for animal's beddings. Also included are normal saline, formalin and all other chemicals were of analytical grade.

Care of rats: Twenty-five adult male Wistar rats of comparable weights (100 ± 20g) were used for this study. The rats were obtained from the animal house of the College of Medicine, Ambrose Alli University, Ekpoma and transferred to the Department of Physiology animal holding facility in the same institution. They were housed in plastic cage with a wire screen top. The animals were fed *ad libitum* with clean tap water and pelleted rat chow during acclimatisation (7 days) and were maintained under standard care of laboratory animals as stated in the guidelines and regulations for the use of animals in scientific experiment.

Preparation of the different colour seeded maize: A bowl each of white and yellow seeded maize were obtained and wash with clean water separately. Thereafter, they were put in iron pot and cooked for 120 to 150 minutes adding adequate amount of water (4.5 to 6 litres) to get them cooked. After cooking, they were allowed to cool and then air dried. After drying they were grinded using electronic machine into coarsely fine powder and were used for the diets formulations.

Diet formulations: The preparation of the 25% and 50% white and yellow seeded maize diets was as described by Nwaopara *et al.* [8]. To prepare 3,000g of the 25% of the white or yellow seeded maize diet, 750g of the grinded white or yellow seeded maize was mixed with 2,250g of normal rat chow and for the 50% white or yellow seeded maize, 1500g of the grinded white or yellow seeded maize was mixed with 1500g of normal rat chow. These were thoroughly mixed by adding adequate clean water to form paste. The pastes were cut into small pieces (approximately 3-5mm diameter) and allowed to dry under the sun. The sun-dried white or yellow seeded maize mixture was then packed into clean sack bag for use.

Experimental grouping: The rats were assigned to five (5) groups of five rats each and treated as follows.

Group 1 (control group): fed normal rat chow (100g daily) and clean water.

Group 2: fed formulated 25% white seeded maize diet (100g daily).

Group 3: fed formulated 50% white seeded maize diet (100g daily).

Group 4: fed formulated 25% yellow seeded maize diet (100g daily).

Group 5: fed formulated 50% yellow seeded maize diet (100g daily).

Duration of feeding: Rats in all the groups received *ad libitum* the different white or yellow seeded maize diets based on corresponding diet group. They were given the different diets daily between the hours of 8:00 am and 10:00 am. Next day leftover diets and water were discarded while rats beddings were changed every other day. The administration of

the different maize diets lasted for 21 days and all animals had access to clean water throughout the treatment period.

Sample collection: At the end of the treatments (22nd day of diet administration), the rats were cervically decapitated and blood sample collected following standard laboratory procedures. The rats were dissected along the lateral side of the neck to assess the jugular blood vessels. Blood samples were then obtained using 2mL syringe and the blood collected into EDTA bottles.

Sample analysis: The blood samples in EDTA bottles were used for the analysis of pack cell volume (PCV), haemoglobin (Hb), red blood cell (RBC), white blood cell (WBC) and differentials. Full Automatic Blood Cell Counter (PCE-210N Ver. 5.42N) was used for assaying the different blood components.

Data analysis: The mean ± standard error of mean was determined and the one-way ANOVA statistical test was performed using SPSS version 20 software. The value $P < 0.05$ was considered statistically significant.

3. RESULTS

Red blood cell markers in rats fed white or yellow seeded maize

Figure 1 compares PCV, Hb and RBC in adult Wistar rats fed white or yellow seeded maize with control diet. PCV decreases with the ingestion of white or yellow seeded maize with statistical decreased in 50% white seeded and 25% and 50% yellow seeded-maize groups compared with the control diet. There was no significant difference ($p>0.05$) in the level of haemoglobin following ingestion of white or yellow seeded maize compared to control diet. However, the ingestion of white or yellow seeded-maize resulted in a decrease in RBC with statistical significance in the 25% and 50% yellow seeded-maize compared with the control.

White blood cell and platelet counts in rats fed white or yellow seeded-maize

Figure 2 compares WBC and platelet count in adult Wistar rats fed white or yellow seeded maize. Although the ingestion of white or yellow seeded maize was likely to decrease WBC, the difference was not significant compared with the control diet. On the other hand, the ingestion of white or yellow seeded maize was likely to increased platelet count, the different was not significant compared with control.

4. DISCUSSION

The findings of this study on the hematological responses of adult male Wistar rats to diets containing different maize varieties offer valuable insights, while also prompting further inquiry. Notably, the reduction in PCV and RBC count observed in rats consuming maize-based diets particularly those fed yellow maize indicates a possible anemic effect. This observation diverges from some prior studies, which reported that properly fortified maize diets can maintain normal hematological parameters in rats [9]. Such differences may stem from variations in the nutrient composition of the maize types used or the lack of dietary supplementation in the present diet of study. Although yellow maize is known for its high carotenoid and provitamin A content [2], the bioavailability of these compounds and their interplay with other essential nutrients for erythropoiesis, such as iron and vitamin B12, is intricate [10]. Consequently, a diet heavily dependent on maize without adequate protein and micronutrient balance may result in deficiencies that impair red blood cell production.

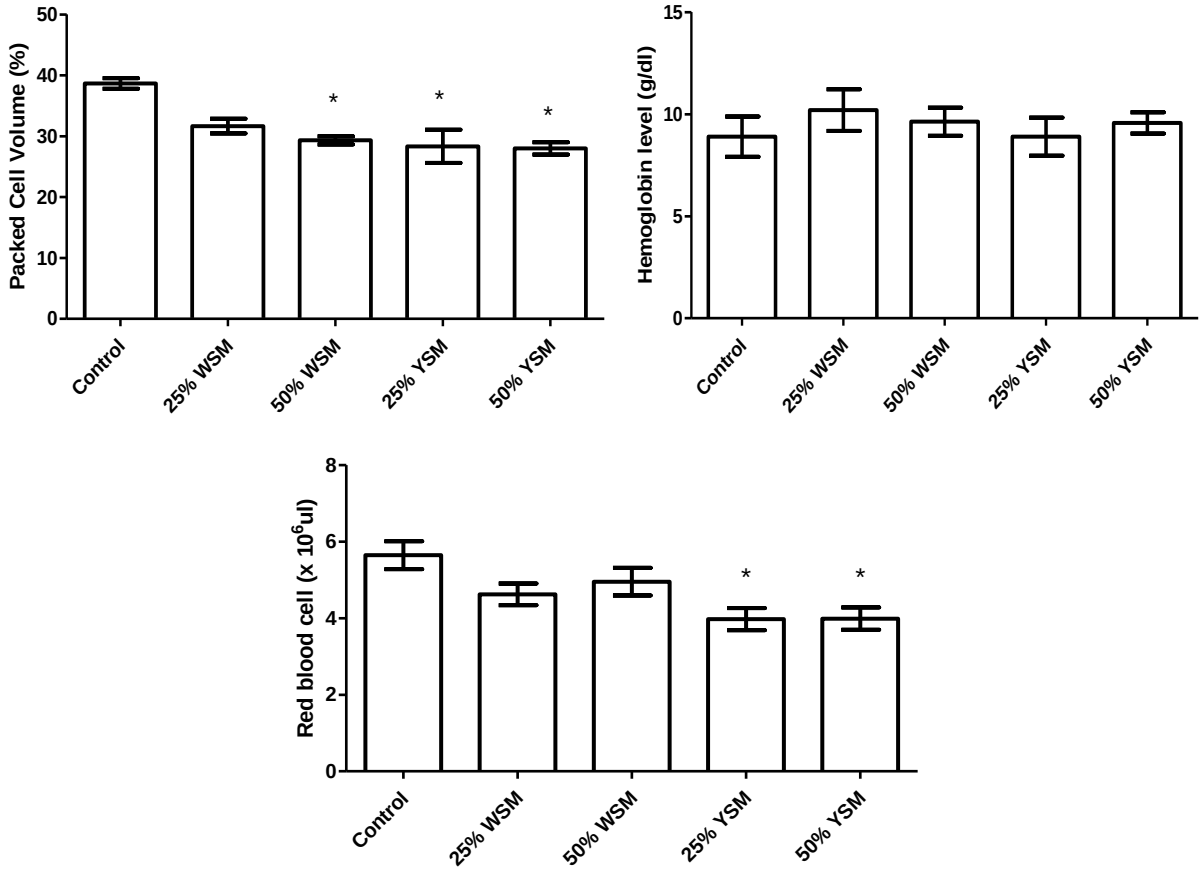


Figure 1: Comparative effect on (a) PCV, (b) Hb and (c) RBC in rats fed white or yellow seeded-maize

Values are means±SEM. * indicate significance ($p<0.05$) compared to the control diet. Key: 25% WSM = 25% of white seeded maize, 50% WSM = 50 % of white seeded maize, 25% YSM = 25% of yellow seeded maize, 50% YSM = 50% of yellow seeded maize

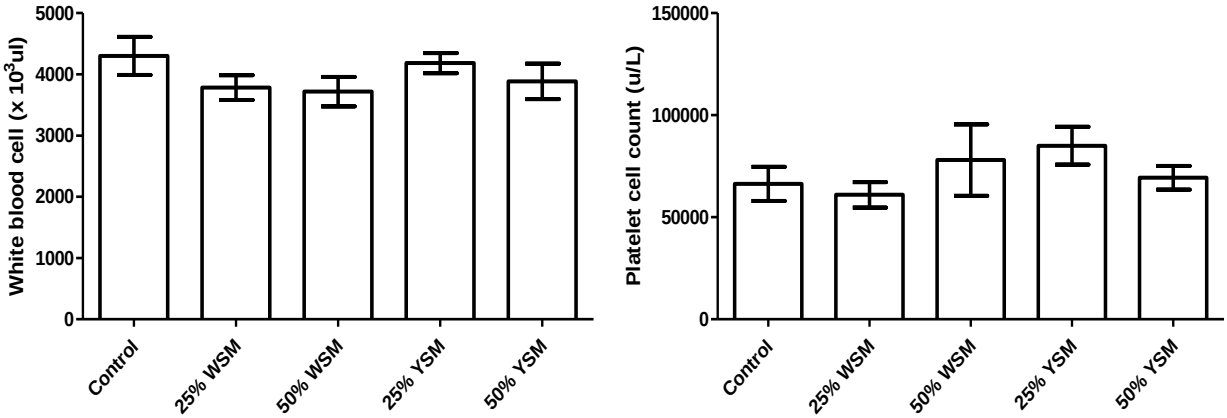


Figure 2: Comparative effect on (a) WBC and (b) platelet count in rats fed white or yellow seeded-maize

Values are means±SEM, No significant difference compared to control diet. Key: 25% WSM = 25% of white seeded maize, 50% WSM = 50 % of white seeded maize, 25% YSM = 25% of yellow seeded maize, 50% YSM = 50% of yellow seeded maize.

Interestingly, this study revealed a decrease in PCV and RBC count without a corresponding significant change in Hb concentration. This finding is unusual, as these hematological parameters typically show a strong correlation. The dissociation observed may suggest a compensatory adaptation, where the remaining red blood cells are either larger in size (macrocytic) or possess higher hemoglobin content (hyperchromic) to preserve the blood's oxygen-carrying capacity [11]. This underscores the importance of conducting more comprehensive hematological evaluations, including measurements of mean corpuscular volume (MCV), mean corpuscular hemoglobin (MCH), and mean corpuscular hemoglobin concentration (MCHC), to better characterize red blood cell morphology. Similar research has documented varied effects of different diets on these parameters; for instance, a study on rats fed cassava, another staple food, reported decreases in RBC and PCV accompanied by a decline in Hb levels [12]. Therefore, the maintenance of stable Hb despite reductions in RBC and PCV in this study is particularly notable.

With respect to immune function, WBC counts remained consistent across all experimental groups, indicating that neither the white nor yellow seeded-maize diets triggered an inflammatory response or adversely affected the immune status of the rats. This finding aligns with previous studies demonstrating that staple food crops generally exert neutral or even supportive effects on immune cell profiles [13]. Additionally, the observed stability in platelet counts is noteworthy, as platelets play a crucial role in blood coagulation. This suggests that the diets did not disrupt thrombopoiesis or enhance platelet destruction, highlighting an important aspect of their safety.

CONCLUSION

This study compelling evidence suggesting a potential anemic effect associated with maize-based diets, particularly the yellow seeded-maize. The observed discrepancies between reductions in PCV and RBC, alongside stable Hb levels, indicate a possible compensatory mechanism that warrants further investigation. Comprehensive future research should incorporate dietary supplementation, detailed nutritional profiling of the feeds, and expanded hematological parameters to clarify the underlying mechanisms.

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